Exploring memorization in hidden states of LLMs via math perturbations

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Abstract

This paper investigates whether large language models solve mathematical problems through genuine reasoning or through pattern-matching drawn from memorized training examples, using the internal representations of two openly available models, Qwen3-4B and Gemma 4 E4B-it, as a window into that question. Building on the MATH-Perturb benchmark, which rewrites existing problems into a 'Simple' variant that keeps the original solution method intact and a 'Hard' variant that changes the reasoning the problem actually requires, the study examines 279 matched pairs of Simple and Hard problems in each model. For every pair, we recorded how each model internally represents the problem immediately before it starts to answer, at every depth of processing inside the network, and measured how far apart the Simple and Hard representations sit from one another. To judge whether this distance is meaningful rather than incidental, we compared it against thousands of pairings between unrelated problems, giving each measurement a calibrated baseline rather than relying on the raw numbers alone.

1 INTRODUCTION

Large Language Models (LLMs) have demonstrated remarkably strong performance across a wide range of mathematical reasoning benchmarks, with frontier models achieving accuracy rates that rival or exceed human level performance on standardized problem sets. However, a fundamental question remains as it is still relatively unknown if this performance reflects genuine mathematical reasoning, or whether it is largely a result of memorization and the reproduction of solution patterns during training.

This distinction carries significant implications. The train ing data for modern LLMs is vast, often containing large portions of the internet, academic repositories, and even competition problem archives. As a result, benchmark problems (even those that are designed to be challenging) may overlap with training data. In such cases, high accuracy on those problems becomes an unreliable signal as a model may arrive to the correct answer, however only doing so through the recognition of surface level structural patterns that they have derived from coming across similar problems in training.

The problem is not merely one of evaluation validity as if LLMs are reliant on memorized patterns rather than an actual possession of mathematical understanding, their reasoning capabilities are inherently brittle, and thus they may fail when problems deviate even slightly from familiar formulations.

To probe this distinction, recent work has introduced *perturbation* based benchmarks. A notable example of this is MATH-perturb which constructs modified variants of existing benchmark problems at two different levels of perturbation. These consist of *simple* perturbations, which still preserve the underlying reasoning conditions of the solutions and the method of solving; as well as *hard* perturbations which fundamentally change the nature of the problem so that the original steps of the problem do not apply whilst still preserving some key characteristics of the original.

LLMs show strong performance on mathematical reasoning benchmarks, however it is largely unclear if this performance comes from true reasoning or through the usage of memorization of previously seen or similar problems and solutions. Many benchmark math problems might overlap with training data, thus high accuracy alone does not prove to be sufficient enough to display that models have learned abstract math concepts, as they could be using surface level solution patterns that they have memorized, rather than an application of the concept.

In an attempt to address this problem, recent benchmarks such as MATH-Perturb insert controlled perturbations of mathematical problems; these consist of *simple* perturbations, which still preserve the underlying reasoning conditions of the solutions and the method of solving; as well as *hard* perturbations which fundamentally change the nature of the problem so that the original steps of the problem do not apply whilst still preserving some key characteristics of the original. Performance degradation under hard perturbation, relative to simple perturbation or the original, is taken as an indicator of over-reliance on memorized solution strategies.

While behavioral evidence from perturbation benchmarks provides indirect evidence of memorization, the internal computational mechanisms underlying this behavior remain poorly understood. It is not yet clear how the internal representations of an LLM change when the model encounters a perturbed variant of a problem it has likely seen before, versus a genuinely novel problem. Do specific layers or attention heads behave differently? Do activation patterns shift in detectable ways? These questions cannot be answered from output accuracy alone; they require looking *inside* the model.

How does internal model behavior differ between original and perturbed math problems? Which aspects of the internal representations (layers, attention heads, and activation patterns) change and how? Can memorization be predicted upstream?

2 RELATED WORKS

High benchmark performance in large language models cannot be taken at face value, as training datasets spanning vast portions of the internet, academic repositories, and competition archives mean that benchmark problems frequently overlap with training data, reducing accuracy to an unreliable signal of genuine ability. When models rely on surface-level pattern recognition rather than true mathematical understanding, their reasoning becomes inherently brittle, failing even under slight deviations from familiar problem formulations. Yet while perturbation benchmarks provide indirect behavioral evidence of this memorization, the internal computational mechanisms remain poorly understood.

There is some recent work in this space. In recent work by Wei Xie et al. [2], the authors modified Cognitive Reflection Test problems to evaluate whether LLMs possess genuine mathematical reasoning, finding that even advanced models like o1 experienced up to a 50% drop in accuracy. The results suggest LLMs rely primarily on pattern matching resembling intuitive System 1 thinking rather than deliberate System 2 reasoning, challenging optimistic views on their progress toward Artificial General Intelligence. However, it only focuses on performance metrics, rather than insights from hidden-state modeling.

Additionally, in the paper "On Memorization of Large Language Models in Logical Reasoning," Chulin Xie et al. [1] investigated model accuracy with regards to their performance on logic puzzles. Through using dynamically generated Knights and Knaves puzzles, the study found that while LLMs can memorize training problems to near-perfect accuracy, they struggle with slight variations, suggesting benchmark saturation may partly reflect memorization. This differs significantly from the research idea presented here, as it looks at general reasoning rather than a model's ability to reason mathematically.

Furthermore, Zhang et al.'s 2024 paper [3] introduces GSM1k, a new benchmark mirroring the style of the widely-used GSM8k, finding accuracy drops of up to 8% in leading LLMs alongside evidence that higher memorization of GSM8k correlates with larger performance gaps, suggesting dataset contamination. Nevertheless, frontier models show minimal overfitting and broadly generalize to novel problems, indicating that while memorization is a concern, it does not fully account for LLMs' mathematical reasoning performance. This diverges from the current paper's idea, as it looks at a potential reason for limitations in reasoning, rather than looking at hidden-state analysis with perturbed problems.

3 METHODOLOGY

The methodology was designed as a controlled, matched-pair analysis of how a mathematical perturbation is represented across the depth of a causal language model. The present analysis combines behavioral screening of three representative outcomes with a pipeline-validation hidden-state study for the C→C example. Its purpose is to establish a defensible outcome taxonomy and demonstrate that activation extraction, vector summarization, geometric comparison, and visualization operate coherently before those measurements are extended to the failure cases.

3.1 Dataset and matched design

MATH-P-Simple and MATH-P-Hard are separate perturbations of the same unreleased original level-5 MATH problem; Hard is not derived from Simple, so the two variants are siblings under a common source rather than one problem rewritten twice. Records were joined by `problem_id`, yielding 279 unique matched pairs, each present in both the Simple and Hard corpora. Each prompt was analyzed before generation so decoded answer tokens could not define the hidden-state distance: the geometry reported throughout this paper describes how the model represents the problem at the moment generation begins, not how it later resolves that problem in text. This ordering matters because any distance computed after decoding would be partly a description of the model's own output rather than of the prompt it was given.

The two checkpoints were chosen to provide a modest generality check without expanding the study beyond what could be run and manually audited on a single machine. Qwen3-4B and Gemma 4 E4B-it come from different model families, use different tokenizers and normalization schemes, and differ in depth and hidden width, so a pattern that appears in only one of them is more likely to reflect an idiosyncrasy of that architecture than a property of matched-perturbation prompts in general. Both are openly available checkpoints, which keeps the exact weights used in this analysis reproducible by a third party rather than dependent on a private or rapidly changing API-served model.

3.2 Checkpoints, templates, and precision

For Qwen3, `enable_thinking=False` changed the rendered string and inserted an empty `<think>...</think>` prefix before answer generation; the checkpoint default matched `enable_thinking=True`. For Gemma, `enable_thinking=False` equaled the default rendering and contained no `<|think|>`; `enable_thinking=True` inserted Gemma's marker. Thus thinking control was verified from rendered prompts rather than inferred from a keyword: relying on the flag's name alone

Table 1: Checkpoint configuration for Qwen3-4B and Gemma 4 E4B-it.

Component	**Qwen3-4B**	**Gemma 4 E4B-it**
Checkpoint	`Qwen/Qwen3-4B`	`google/gemma-4-E4B-it`
Sites × width	$37 \times 2{,}560$	$43 \times 2{,}560$
Forward precision	bfloat16	bfloat16
Stored activations	float16	float16
Final block hook	`model.layers.35`	`language_model.layers.41`
Final site	post-final norm, site 36	post-final norm, site 42

would have been unsafe, since the same setting toggles opposite default behaviors in the two checkpoints, and a script that assumed symmetry across models could have silently compared a Qwen3 thinking prompt against a Gemma non-thinking prompt under the same nominal setting. Qwen's older 279-pair Qwen2.5-Math-7B-Instruct artifacts are a different checkpoint and are not used for any Qwen3 claim; keeping this boundary explicit avoids conflating results from two model families that happen to share a similarly sized matched-pair count.

3.3 Activation extraction and pooling

All 558 prompts per model were forwarded individually with `output_hidden_states=True`, no generation, and a 1,024-token input limit. At each returned site we retained the last prompt token, the mean of the final three prompt tokens, and the mean across valid prompt tokens; these three pooling rules were chosen to span a range from maximally local, a single boundary token, to maximally global, the full prompt, so that any separation result could be checked against the possibility that it depended on one arbitrary choice of summary statistic. Separate forward hooks captured the true final transformer-block output before final normalization. Arrays were stored as float16 and metrics computed after conversion to float32; bfloat16 forward and float16 storage may suppress very small differences, a limitation that matters most for pairs whose true distance is close to zero.

$$r_{\mathrm{rel}}(a,b) = \frac{a - b_2}{\frac{1}{2}\,(a_2 + b_2)} \tag{1}$$

$$\cos(a,b) = \frac{a \cdot b}{a_2\, b_2} \tag{2}$$

Collecting hidden states without generation was a deliberate simplification, not only a computational convenience. Running a full forward pass with no sampling and no continuation is deterministic and roughly two orders of magnitude cheaper than generating a complete solution, which made it feasible to extract activations for every one of the 558 prompts per model at every representation site and under three pooling rules, rather than restricting the geometric analysis to a small hand-picked subset. The trade-off is that this approach can only describe how a model represents a problem before it starts to answer it, not how that representation evolves while the model is actually reasoning through a solution; extending the method to generation-time states is noted as a direction for future work rather than attempted here.

3.4 Null distributions and confound model

For each pooling rule and site convention we computed the 279 matched Simple$_i$–Hard$_i$ distances, 2,000 fixed-seed mismatched Simple$_i$–Hard$_j$ distances with $i \neq j$, and 2,000 mismatched within-Simple distances. The within-Simple set serves as a second null: because every Simple-Hard cross comparison mixes both a different source problem and a different perturbation type, comparing against within-Simple mismatches isolates how much of the separation is attributable to problem identity alone, holding perturbation type fixed. AUC is the probability that a randomly drawn matched pair is closer than a randomly drawn mismatched cross-type pair, estimated directly from these three sets of distances rather than from a fitted curve. For matched pairs, we regressed at the final-normalized last token on absolute prompt-token-count difference and normalized Levenshtein distance to test how much of the separation textual form alone could explain.

The confound model was deliberately kept simple. A two-covariate linear regression cannot capture nonlinear or interaction effects between token-count difference and edit distance, and it treats every matched pair as an independent observation despite pairs sharing an underlying problem source with other statistics reported elsewhere in this paper. The regression is used here only as a screening tool, to ask whether textual form alone could plausibly account for the separation reported in Section 3.1, not as a finished causal model of what drives prompt-boundary distance.

3.5 Validation

Every activation array was finite and had its expected shape. Across all 279 pairs in both models, last-token site-0 relative L2 was exactly zero, which is expected because the final prompt token before generation is a fixed template token shared across every Simple/Hard pair within a model, and this check confirms token alignment rather than constituting a finding in its own right. Seeds were fixed at `torch.manual_seed(0)` and `numpy.random.default_rng(0)`, and each analysis wrote its library versions, so that a future rerun can be checked against the exact software environment used here rather than against a nominal package list alone.

4 RESULTS

4.1 Null calibration of prompt-boundary distance

Matched prompts were substantially closer than unrelated prompts in both models. This is the principal positive result: prompt-boundary geometry carries strong information about which Simple and Hard prompts belong to the same source problem. Concretely, a randomly drawn matched pair was closer than a randomly drawn mismatched pair in 95.2–98.6% of draws across sites and models (Table 2). Figure 1 shows this separation directly: the matched-pair distribution sits as a narrow peak near zero in both panels, while the mismatched distribution forms a broader, right-shifted mass. The overlap between the two distributions is visibly larger for Gemma than for Qwen3, consistent with Gemma's lower last-block AUC (0.952) in Table 2 and with the wider spread evident in its final-normalized distances.

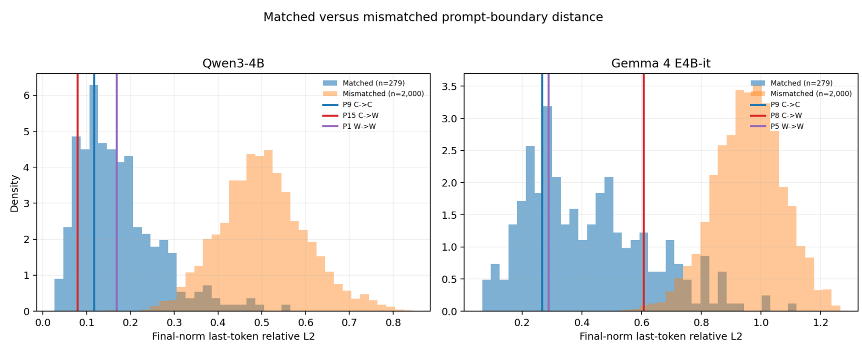

Figure 1: Final-normalized last-token relative L2 for matched and mismatched prompts. Vertical lines mark the six selected cases; each panel has its own case legend.

Table 2: Null calibration of final-normalized last-token relative L2, by model and site convention.

Model	Site	Mean	SD	P5	P50	P95	AUC
Qwen3-4B	last block	0.206	0.110	0.072	0.183	0.428	0.986
Qwen3-4B	final norm	0.171	0.092	0.059	0.152	0.354	0.986
Gemma 4 E4B-it	last block	0.211	0.115	0.067	0.192	0.428	0.952
Gemma 4 E4B-it	final norm	0.421	0.206	0.154	0.383	0.813	0.980

4.2 Pooling-rule sensitivity

Table 3 repeats the null calibration under two alternative pooling rules. Separation held under every pooling choice tested, with AUC never falling below 0.980 for either model, indicating that the matched-pair signal is not an artifact of the last-token pooling rule used elsewhere in this paper. Prompt-mean pooling produced the highest AUC for both models, 0.996 for Qwen3 and 0.998 for Gemma, ahead of last-three-token and last-token pooling in that order. This ordering is consistent with a signal that is distributed across the prompt rather than concentrated at the boundary: averaging over every valid prompt token

Table 3: Null calibration under alternative pooling rules.

Model	Pooling	Mean	P25	P50	P75	AUC
Qwen3-4B	last_token	0.171	0.106	0.152	0.212	0.986
Qwen3-4B	last_three_mean	0.121	0.079	0.107	0.155	0.987
Qwen3-4B	prompt_mean	0.129	0.069	0.102	0.167	0.996
Gemma 4 E4B-it	last_token	0.421	0.266	0.383	0.546	0.980
Gemma 4 E4B-it	last_three_mean	0.412	0.286	0.412	0.515	0.984
Gemma 4 E4B-it	prompt_mean	0.172	0.105	0.152	0.219	0.998

dilutes any single-token noise while retaining the aggregate difference between a Simple and a Hard restatement of the same problem, whereas last-token pooling depends on whichever content the perturbation happened to place immediately before generation.

4.3 Case percentiles and exact site separation

Each row in Table 4 reports a single selected observation, not a sample. The matched and cross percentiles show where that pair's final-normalized distance falls within the 279 matched pairs and the 2,000 mismatched cross-type draws, respectively; for relative L2, a higher percentile indicates greater representational separation.

Table 4: Case percentiles and exact site separation for the six selected cases.

Model	Label	ID	Last block d	Final norm d	Matched pct.	Cross pct.	Norm ratio
Qwen3-4B	C→C	9	0.142	0.116	31.2%	0.0%	1.015
Qwen3-4B	C→W	15	0.097	0.079	11.8%	0.0%	0.997
Qwen3-4B	W→W	1	0.195	0.168	57.0%	0.1%	0.976
Gemma	C→C	9	0.122	0.266	24.7%	0.0%	0.986
Gemma	C→W	8	0.374	0.607	80.6%	0.2%	0.862
Gemma	W→W	5	0.152	0.288	31.5%	0.0%	0.945

Final normalization materially changes the reported distances, especially for Gemma problem 8, where relative L2 rises from 0.374 before normalization to 0.607 after. The accompanying Simple/Hard norm ratio of 0.862 shows a magnitude change at the same boundary, but because relative L2 is scale-normalized, rescaling alone cannot fully explain the shift. The strongest claims therefore attach to explicit site conventions, not an ambiguous "final layer." This late-layer amplification is visible directly in Figure 6, where the Gemma problem 8 trajectory stays under roughly 0.15 relative L2 through the first thirty sites before rising sharply toward the final few layers, so the case's headline distance depends heavily on exactly which of those final sites is reported.

4.4 Textual confounds

Normalized textual edit distance carries a positive coefficient in both models. The two covariates jointly explain 35.2% of Qwen and 24.3% of Gemma

Table 5: Regression of matched-pair relative L2 on textual confounds.

Model	**Intercept**	$\|\boldsymbol{\Delta}$ tokens$\|$	**Normalized edit distance**	R^2
Qwen3-4B	0.119	−0.00017	0.357	0.352
Gemma 4 E4B-it	0.324	−0.00087	0.692	0.243

matched-pair variance: substantial confounding, but not enough to reduce prompt-boundary distance to text length and edit distance alone. The near-zero magnitude of the token-count coefficient relative to the edit-distance coefficient in both models suggests that it is the substance of what changed in the prompt, not merely how much longer or shorter it became, that drives most of the explained variance. Residual percentiles, after removing this textual component, place Qwen problems 9, 15, and 1 at 43.0%, 26.2%, and 75.3%, and Gemma problems 9, 8, and 5 at 34.1%, 90.0%, and 41.9%.

4.5 Saved case studies

4.5.1 Qwen3-4B: C→C, problem 9

Both answers were correct; the pair nonetheless registers measurable representational divergence (Table 4), showing that correctness and distance are not mutually exclusive. Figure 2 shows this divergence emerging gradually rather than at a single layer: the last-token trajectory tracks close to the other two pooling curves through roughly site 20, then separates and climbs to just above 0.10 relative L2 by the final block, while prompt-mean pooling stays essentially flat near 0.02–0.03 across the full depth of the network.

4.5.2 Qwen3-4B: C→W, problem 15

The Simple answer, 125°, was manually adjudicated equivalent to 125 after the parser missed the degree symbol; Hard returned 75 rather than the correct 43. The C→W label therefore rests on manual normalization and remains provisional pending the parser audit. As in problem 9, Figure 3 shows the last-token distance separating from the pooled trajectories only in the second half of the network, reaching roughly 0.07–0.08 relative L2 by the final site, a magnitude comparable to the C→C case despite the differing correctness outcome.

4.5.3 Qwen3-4B: W→W, problem 1

Simple ended without a valid boxed answer, and Hard returned 16 rather than the correct 36; this mixed completion/error failure illustrates the heterogeneity folded into the single W→W label. Figure 4 shows a markedly larger final separation than either Qwen case above, with the last-token trajectory reaching roughly 0.15–0.16 relative L2, consistent with problem 1's high matched-pair percentile of 57.0% in Table 4.

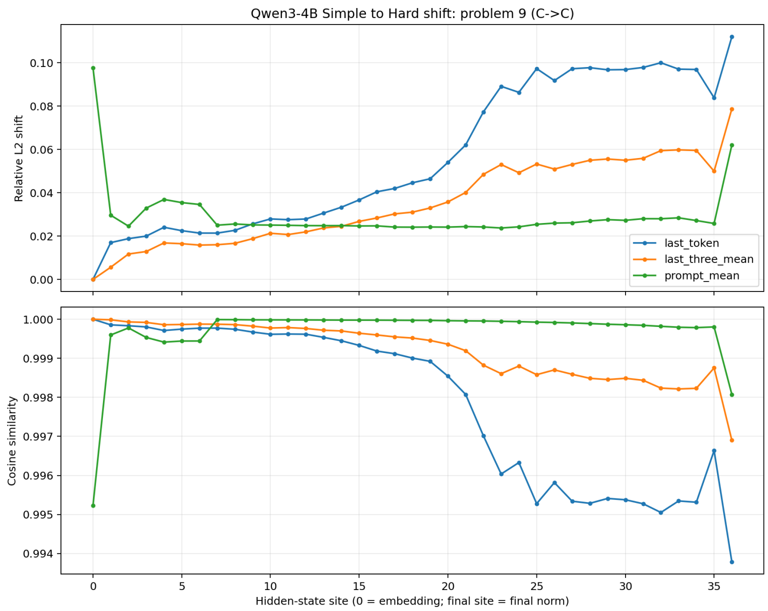

Figure 2: Qwen3-4B problem 9 (C→C): layerwise relative L2 and cosine trajectories for three pooling rules.

4.5.4 Gemma 4 E4B-it: C→C, problem 9

Both answers were correct, yet last-three-token pooling distance exceeds last-token distance. This inversion was reproduced under fresh token-level recomputation, after checking the token axis, slice, padding, and final-token identity. Figure 5 shows the inversion is driven by a sharp, transient spike in the last-three-token trajectory around site 20, where relative L2 climbs to roughly 0.4 and cosine similarity drops to about 0.92 before partially recovering; the last-token trajectory stays comparatively smooth across the same span, only rising toward the final sites.

4.5.5 Gemma 4 E4B-it: C→W, problem 8

Simple was correct; Hard returned 33 rather than the correct 35. The handshake method remains applicable, so the failure combines an arithmetic error with a lapse in enforcing the odd-degree parity constraint, making this R2 rather than R1 under the proposed reuse rubric. The failure is concentrated almost entirely in the final few sites: cosine similarity holds above 0.95 through most of the network before falling sharply to roughly 0.82 at the last recorded site, mirroring the late-layer rise already noted for this case's relative L2 trajectory.

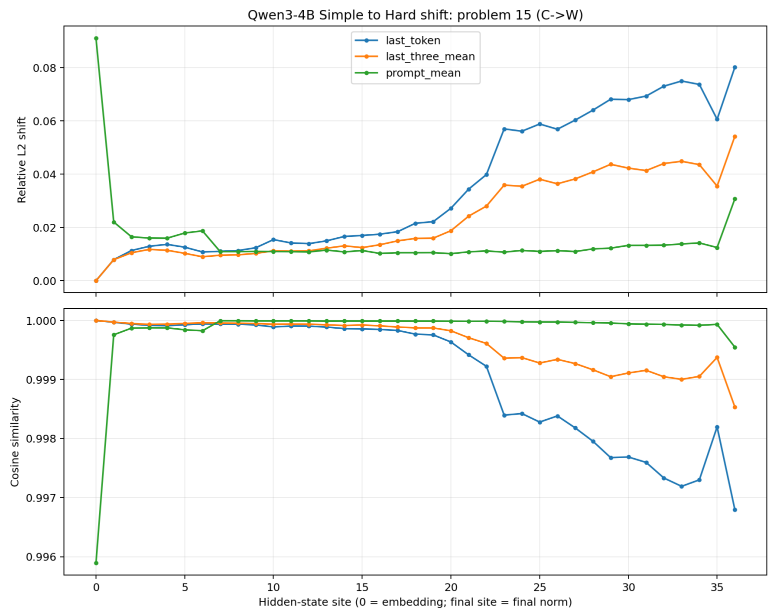

Figure 3: Qwen3-4B problem 15 (C→W): layerwise relative L2 and cosine trajectories for three pooling rules.

4.5.6 Gemma 4 E4B-it: W→W, problem 5

Both boxed answers were mathematically wrong on a parabola-line tangency task. Selection was deterministic: the lowest `problem_id` among complete-box W→W candidates after manual verification. Figure 7 shows a more gradual divergence than problem 8's late spike or problem 9's mid-network spike: relative L2 rises steadily from around site 15 onward across all three pooling rules, reaching approximately 0.27 at the final last-token site, a magnitude close to problem 9's own final value but reached through steady accumulation rather than a single transient event, and well below problem 8's late-layer surge past 0.6.

Read together, the six cases do not tell a single story, and that is itself informative. Both correct-to-correct cases, Qwen3 problem 9 and Gemma problem 9, show measurable late-layer separation despite full behavioral success, confirming that internal divergence is not on its own a warning sign. The two correct-to-wrong cases differ sharply in shape: Qwen3's separation develops gradually across the second half of the network, while Gemma's is concentrated almost entirely in the final few sites, suggesting that a Hard-specific breakdown is not a single mechanism but at least two distinct ones occurring at different depths. The two wrong-to-wrong cases are the least alike of all: one reflects an incomplete generation rather than a mathematical error, the other a clear but wrong computation, and pooling them under one label, as the behavioral screen

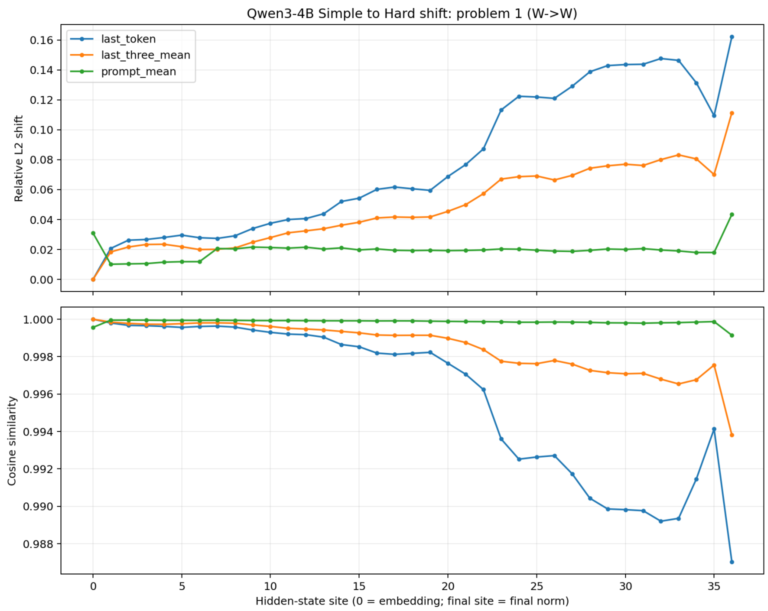

Figure 4: Qwen3-4B problem 1 (W→W): layerwise relative L2 and cosine trajectories for three pooling rules.

currently does, visibly discards this distinction. No pattern here is common to more than one or two cases at a time, which is consistent with the paper's central caution: six illustrative examples are not a sample, and any apparent regularity across them should be treated as a hypothesis for the balanced follow-up study to test, not as a conclusion in its own right.

4.6 Behavioral-screen diagnostic

The original Gemma screen is a pipeline failure, not a behavioral finding. Of 558 responses, 433 (77.6%) reached exactly 512 tokens; 249 of 279 pairs reached the cap on at least one side, and 248 of 258 NO_BOX pairs included a capped response. The 512-token screen therefore cannot support Gemma accuracy or transition prevalence. Figure 8 makes the scale of the failure visible directly: the generated-token histogram is dominated by a single tall bar sitting exactly at the 512-token cap, with only a modest cluster of short, genuinely completed responses below roughly 50 tokens and almost nothing in between. A screen whose output distribution looks like this is measuring where generation was cut off, not what the model would have produced given room to finish.

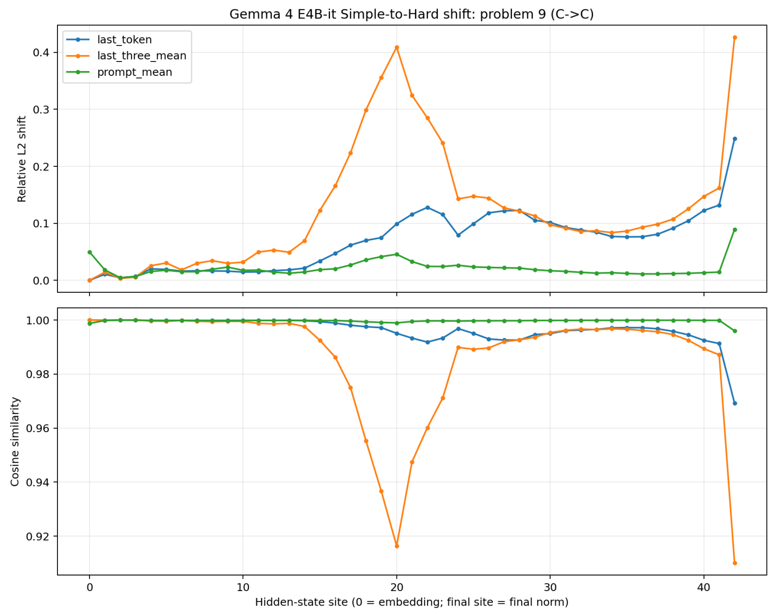

Figure 5: Gemma 4 E4B-it problem 9 (C→C): layerwise relative L2 and cosine trajectories for three pooling rules.

4.6.1 Partial 2,048-token pilot

A fixed-seed random pilot was stopped after 14 of 20 pairs at the user's request. Among 28 completed responses, 2 reached 2,048 tokens (7.1%); 11 of 14 pairs were still NO_BOX. Because the pilot is incomplete and exceeds the work order's 5% cap gate in its observed subset, it is reported only as a diagnostic and is excluded from accuracy estimates. That 11 of the 14 completed pairs were still unresolved even at four times the original token budget suggests the underlying difficulty is not simply that Gemma runs long, but that a meaningful share of these problems may not have been on track to reach a boxed answer within any bounded generation budget, a possibility the original 512-token screen could not have distinguished from ordinary truncation.

4.6.2 Parser and technique coding status

Reproducible sampling sheets and Wilson-interval/Cohen-κ analysis code were prepared, but no human adjudication was performed. Parser error rates, corrected accuracy bounds, technique-reuse proportions, and inter-rater agreement are therefore not reported. In particular, Gemma problem 8 cannot serve as the proposed R1 exemplar: its counting method remains valid, making it an R2 arithmetic/constraint-execution failure under the supplied definitions.

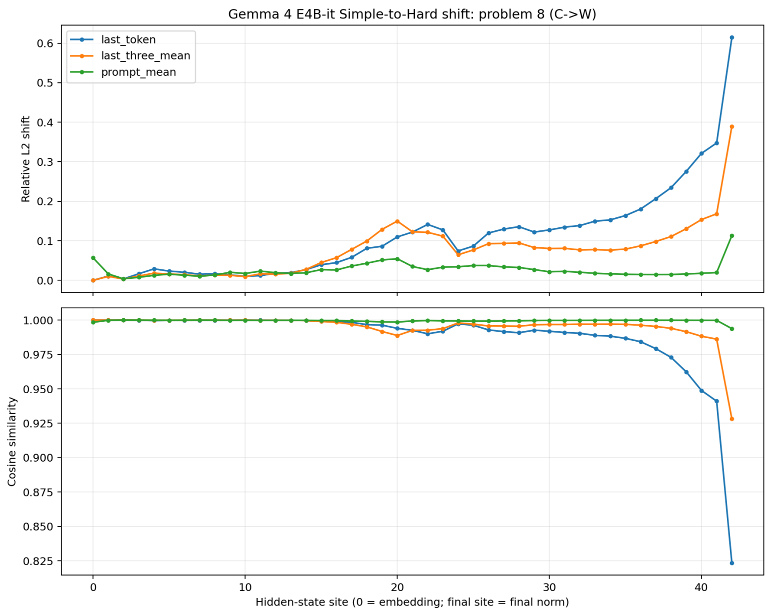

Figure 6: Gemma 4 E4B-it problem 8 (C→W): layerwise relative L2 and cosine trajectories for three pooling rules.

5 DISCUSSION

5.1 What the six cases establish

The cases support existence claims only. Preserved correctness can coexist with divergence; a provisional W→W label can combine distinct proximal failures; and final normalization can amplify a pair's apparent distance. The cases do not estimate C→C, C→W, or W→W central tendencies, and their numerical ordering is not a finding.

5.2 Why the behavioral claims remain withdrawn

The 512-token Gemma corpus is selected for short solutions and early abandonment, and the longer pilot is incomplete. The Qwen3 behavioral corpus has not been rerun under the repaired protocol. Parser validity has not been estimated from random samples. Each of these gaps compounds rather than merely accumulates: a selection bias in which responses survive the token cap interacts with an unvalidated parser in ways that could shift an apparent accuracy figure in either direction, so a partial fix to only one of these three problems would not by itself make a behavioral claim defensible. Consequently, cross-model accuracy, transition prevalence, solution-technique reuse, and memorization claims remain outside the evidence.

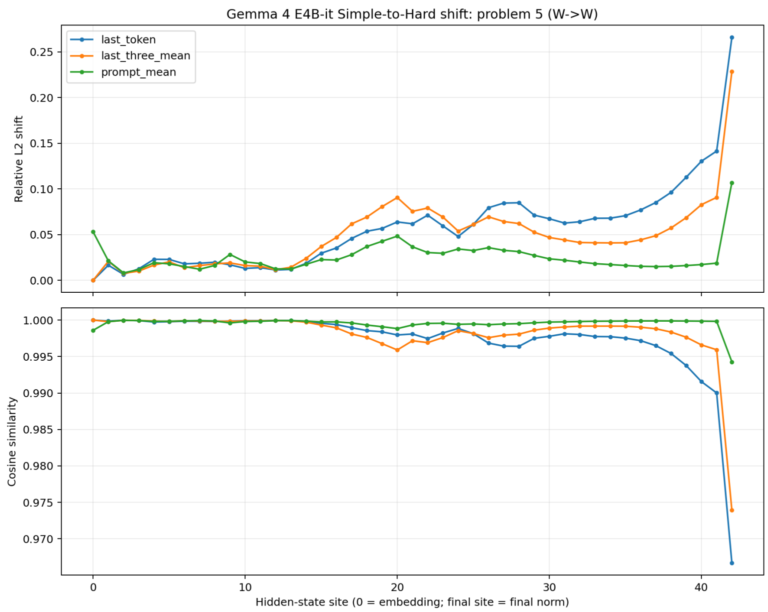

Figure 7: Gemma 4 E4B-it problem 5 (W→W): layerwise relative L2 and cosine trajectories for three pooling rules.

This failure is worth dwelling on beyond its effect on this paper's own results, because it illustrates a risk that applies to perturbation benchmarks generally. MATH-Perturb, and benchmarks like it, are attractive precisely because Hard variants are designed to demand more from a model, which also means they are more likely to produce longer responses. A fixed generation budget chosen without checking its effect on the harder condition will systematically truncate exactly the responses the benchmark was built to examine, biasing any resulting accuracy comparison in a direction that is easy to miss unless the raw generation-length distribution is inspected directly, as it was here.

5.3 Limitations

- The six outcome labels are provisional and contain one selected case per model-cell, so no case in this paper should be read as evidence about how common R1 versus R2 failures, or C→W versus W→W outcomes, are across the broader corpus.
- The null calibration concerns prompts, not generated answers, and is observational rather than causal: it establishes that matched and mismatched pairs are geometrically distinguishable, not that this geometry causes, or is used by the model to produce, any particular downstream answer.

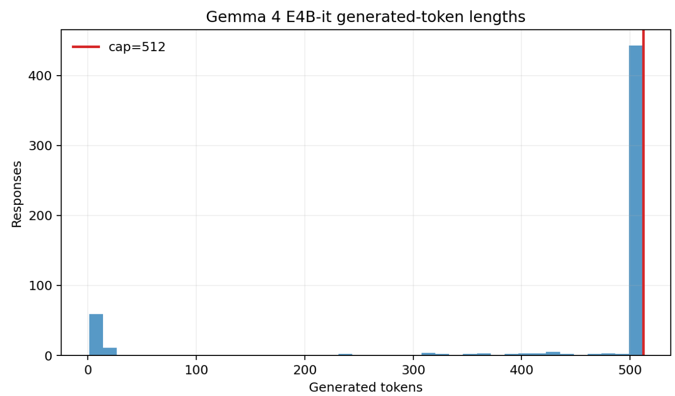

Figure 8: Gemma generated-token distribution under the invalid 512-token screen.

- Qwen and Gemma differ in architecture, normalization, tokenization, precision history, and hardware; raw site indices are not aligned coordinates, so a label such as 'site 20' means something different in each model, and the cross-model site comparisons drawn in the case studies above are illustrative rather than a registered alignment.
- Forward computation used bfloat16 and stored pooled activations as float16 before float32 metric evaluation, a precision path that may compress the very smallest true differences toward zero and could understate separation for near-identical pairs more than for pairs the model already represents as clearly distinct.
- The parser audit, corrected accuracy bounds, technique coding, and full 4,096-token reruns remain incomplete, and until they are finished, none of the percentiles or case labels reported here should be treated as final.
- AUC values are reported as point estimates without confidence intervals or a formal test of whether the separation strength in Qwen3 and Gemma differs by more than sampling variation in the null draws would predict; readers should treat the roughly three-point AUC gap between the two models as suggestive rather than established.

6 CONCLUSION

This study developed an outcome-based, cross-model framework for analyzing matched Simple and Hard mathematics perturbations in Qwen3-4B and Gemma 4 E4B-it, verifying 279 matched dataset pairs per model and examining one representative example from each of three distinct outcome categories (C→C, C→W, and W→W) in both checkpoints. Beyond the six individual cases, prompt-boundary hidden states across the full 279-pair set were compared against a calibrated null distribution of mismatched pairs, showing that a randomly drawn matched pair sat closer than a randomly drawn mismatched pair in the large majority of comparisons in both models (AUC of 0.986 for Qwen3-4B and 0.980 for Gemma). This establishes, at the population level rather than just the six-case level, that representational divergence between Simple and Hard prompts is a real and measurable signal, not an artefact of the specific cases chosen for close reading.

Problem 9 (C$\rightarrow$C) had Qwen3-4B correctly predicting 69 and 48; Gemma's own problem 9 (C$\rightarrow$C) likewise produced two correct answers. Saved hidden states for both cases showed representational divergence developing across the depth of the network despite fully successful behavior, though the two models diverged differently: Qwen3-4B's separation grew steadily through the second half of the network, while Gemma's showed a sharp, transient spike around the network's midpoint before partially recovering. Together these confirm, across two independent architectures, that representational divergence can coexist with successful behavior, providing a necessary control for the failure analysis that follows.

Problem 15 (C$\rightarrow$W, manually validated) produced a Qwen3-4B Simple response of 125° that is mathematically correct despite being rejected by the automated parser, against a Hard prediction of 75 — incorrect relative to the ground truth of 43. Gemma's C$\rightarrow$W case, problem 8, showed a comparable pattern under a different mechanism: its Simple response was correct, but its Hard response of 33 (against a ground truth of 35) reflected a valid handshake-method approach undermined by an arithmetic slip and a lapse in enforcing the odd-degree parity constraint, an arithmetic and constraint-execution failure rather than an abandoned method. Gemma's failure was also concentrated almost entirely in the final few layers, where cosine similarity fell sharply, unlike Qwen3-4B's more gradual late-network separation. Both cases point to Hard-specific breakdowns, but the two models appear to fail through distinct mechanisms rather than a shared one.

Problem 1 (W$\rightarrow$W, under a strict output rule) showed Qwen3-4B's Simple run producing no valid final boxed answer, and its Hard run predicting 16 against a ground truth of 36. Gemma's W$\rightarrow$W case, problem 5, was more straightforwardly wrong on both variants: a parabola-line tangency problem where both boxed answers were mathematically incorrect, with divergence accumulating steadily rather than through a single abrupt shift. The mismatch between Qwen3-4B's two failure modes, and the further contrast with Gemma's more uniform failure, indicates that W$\rightarrow$W cases require further stratification rather than treatment as a single category, regardless of which model produced them.

The comparison also exposes a different evidential boundary than before. Hidden-state activations were ultimately saved for all six cases across both models, so the earlier gap, where only the C$\rightarrow$C example had saved activations, no longer applies. The boundary that remains is behavioral rather than representational: Gemma's accuracy pipeline was found to rest on an invalid 512-token generation cap that truncated most of its responses to the harder problems before completion, meaning Gemma's own correctness labels above should be read as provisionally adjudicated rather than fully validated at scale. The next experiment should rerun Gemma's behavioral scoring under a corrected token budget, then extend the same balanced, parser-audited comparison already applied to the hidden-state trajectories.

The strongest present conclusion is that transition-aware, cross-model analysis is essential. C→C establishes the range of internal change compatible with robustness in two independent architectures; C→W isolates perturbation-specific behavioral loss, arising through different mechanisms in each model; W→W separates broader difficulty from Hard-specific breakdown, and shows that even superficially identical labels can mean different things across models. With manual adjudication, a corrected Gemma behavioral pipeline, balanced samples, uncertainty estimates, and activation patching, this framework can support a rigorous investigation of whether specific internal changes are associated with failure under mathematical perturbation, and whether that relationship holds beyond a single model family.

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